

Ivan Evdokimov, Ph.D.

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Professional Summary

Software Engineer with 4+ years deploying enterprise-scale ML and GenAI solutions into production. Expert in building end-to-end MLOps pipelines with model governance, drift detection, and automated retraining on AWS. Proven track record of working in diverse multi-national teams and taking AI models from experimentation to production, processing sensitive social sciences datasets with 95%+ accuracy while ensuring scalable deployment.

Experience

Software Engineer – GenAI, University of Essex — Colchester, UK Nov 2025 — Present

- Designed and engineered production-grade CI/CD pipelines in Python for AI-powered development workflows, leveraging GitHub Actions and Vercel to enable automated deployment of codebases with 90%+ code generation rate and weekly release cycles, demonstrating hands-on Python expertise and scalable system architecture.
- Built automated testing and quality assurance frameworks in Python for AI-generated code, implementing robust validation pipelines that reduced manual review time by 70% while maintaining production stability, security standards, and code quality.
- Developed and deployed end-to-end MLOps infrastructure in Python for a multi-tenant SaaS platform serving global B2B clients, accelerating feature delivery from monthly to weekly releases through AI-driven development automation, and contributing to engineering standards and technical direction.

Software Engineer – GenAI, UK Data Service — Colchester, UK Jan 2023 — Oct 2025

- Trained, fine-tuned, and deployed production-grade Generative AI models in Python (PyTorch, scikit-learn) using Hugging Face, vLLM, and Ollama for automated metadata classification and disclosure risk assessment, achieving 95%+ accuracy on sensitive social science datasets while applying modern AI patterns including LLM orchestration, retrieval-based approaches, and structured evaluation techniques.
- Architected and implemented serverless ML infrastructure on AWS (Lambda, Step Functions, EFS) with automated model retraining pipelines in Python, reducing manual data validation workload by 50% and enabling scalable, maintainable production systems that empower non-technical users.
- Optimised statistical disclosure control algorithms through performance engineering, migrating from R to C++ with bitmask techniques to achieve 11x performance improvement, enabling real-time processing of 500k+ record datasets while maintaining robust software engineering standards and code quality.

Research Officer & ML Instructor, University of Essex — Colchester, UK Oct 2021 — Dec 2022

- Engineered high-performance computing solutions by migrating quantitative macroeconomic models from MATLAB to C++, reducing simulation runtime by 8x to enable efficient cross-institutional collaboration, demonstrating strong software engineering fundamentals, design, and performance optimisation.
- Delivered technical instruction in Data Structures, C/C++, and Machine Learning to 100+ undergraduate and postgraduate students, instilling engineering discipline, code quality, and maintainability principles.

Projects

Chat-LLM github.com/ivan020/ChatLLM

- Engineered a Python-based self-hosted Ollama MCP server on Raspberry Pi with Twitch chat integration, demonstrating hands-on Generative AI implementation, LLM orchestration, and agentic workflows in a production-like environment.

FinBot: Chat-Based Trading Simulation github.com/ivan020/FinBot

- Developed a real-time trading simulation chatbot in Python with Twitch integration and a PostgreSQL backend API (Supabase), showcasing full-stack engineering skills and production-grade application development.

MTG Card Reader ivan020.github.io/MtgRecognizer/

- Engineered and deployed a Golang REST API with LLM integration for card data mapping against PostgreSQL, applying modern AI patterns, retrieval-based approaches, and structured evaluation techniques in a scalable production service.

Education

University of Essex, PhD in Computational Finance Oct 2021 — May 2025

- Research and developed PyTorch-based transfer learning frameworks for Bayesian model averaging in financial forecasting, publishing peer-reviewed papers and releasing an open-source package adopted externally, demonstrating deep Generative AI expertise and production-ready research implementation.

University of Essex, MSc in Financial Econometrics Oct 2020 — Sep 2021

- Engineered C++ agent-based financial simulations to model macroeconomic dynamics under negative interest rates, applying strong software engineering fundamentals to complex financial modeling problems.